

DTU Physics

Sources and Monitors

part 2.



Sources: Source model overview

➤ **Mathematical:**

- *Source_simple.comp*
- *Source_div.comp*

➤ **Pulsed sources:**

- *ESS_butterfly.comp*
- *ESS_moderator.comp*
- *Moderator.comp*
- *SNS_source.comp (*)*
- *SNS_source_analytic (*)*
- *ViewModISIS (*)*
- *ISIS_moderator.comp (*)*

➤ **Reactors :**

- *Source_Maxwell_3.comp*
- *Source_gen.comp*
- *Source_gen4.comp*
- *Source_multi_surfaces.comp (*)*
- **I/O mechanisms:**
 - ***MCPL_input/output.comp***
 - *Virtual_input/output.comp*
 - *Virtual_mcnp_ss_input/output.comp*
 - *Virtual_tripoli4_input/output.comp*
 - *Vitess_input/output.comp*

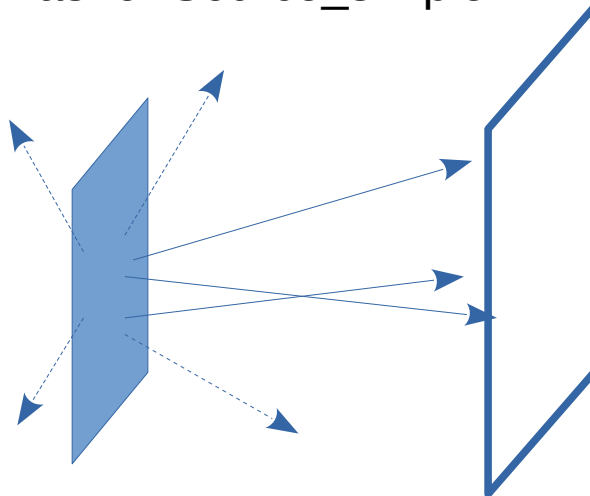


Sources: Source_Maxwell_3

```
COMPONENT source = Source_Maxwell_3(yheight=0.156, xwidth=0.126,  
    Lmin=0.1, Lmax=9.0, dist=1.5, focus_xw = 0.025, focus_yh = 0.12,  
    T1=150.42, I1=3.67E11, T2=38.74, I2=3.64E11, T3=14.84, I3=0.95E11)
```

Parameters from the PSI cold source

Initial position and direction: as for Source_simple





Sources: Source_Maxwell_3

```
COMPONENT source = Source_Maxwell_3(yheight=0.156, xwidth=0.126,
    Lmin=0.1, Lmax=9.0, dist=1.5, focus_xw = 0.025, focus_yh = 0.12,
    T1=150.42, I1=3.67E11, T2=38.74, I2=3.64E11, T3=14.84, I3=0.95E11)
```

Parameters from the PSI cold source

Intensity at a given wavelength drawn from a sum of (up to) 3 normalized Maxwellian distributions:

$$I(\lambda) = \sum_{i \in 1,2,3} I_i M(\lambda, T_i); \quad M(\lambda, T_i) = 2 \alpha^2 \exp\left(\frac{-\alpha}{\lambda^2}\right) / \lambda^5;$$

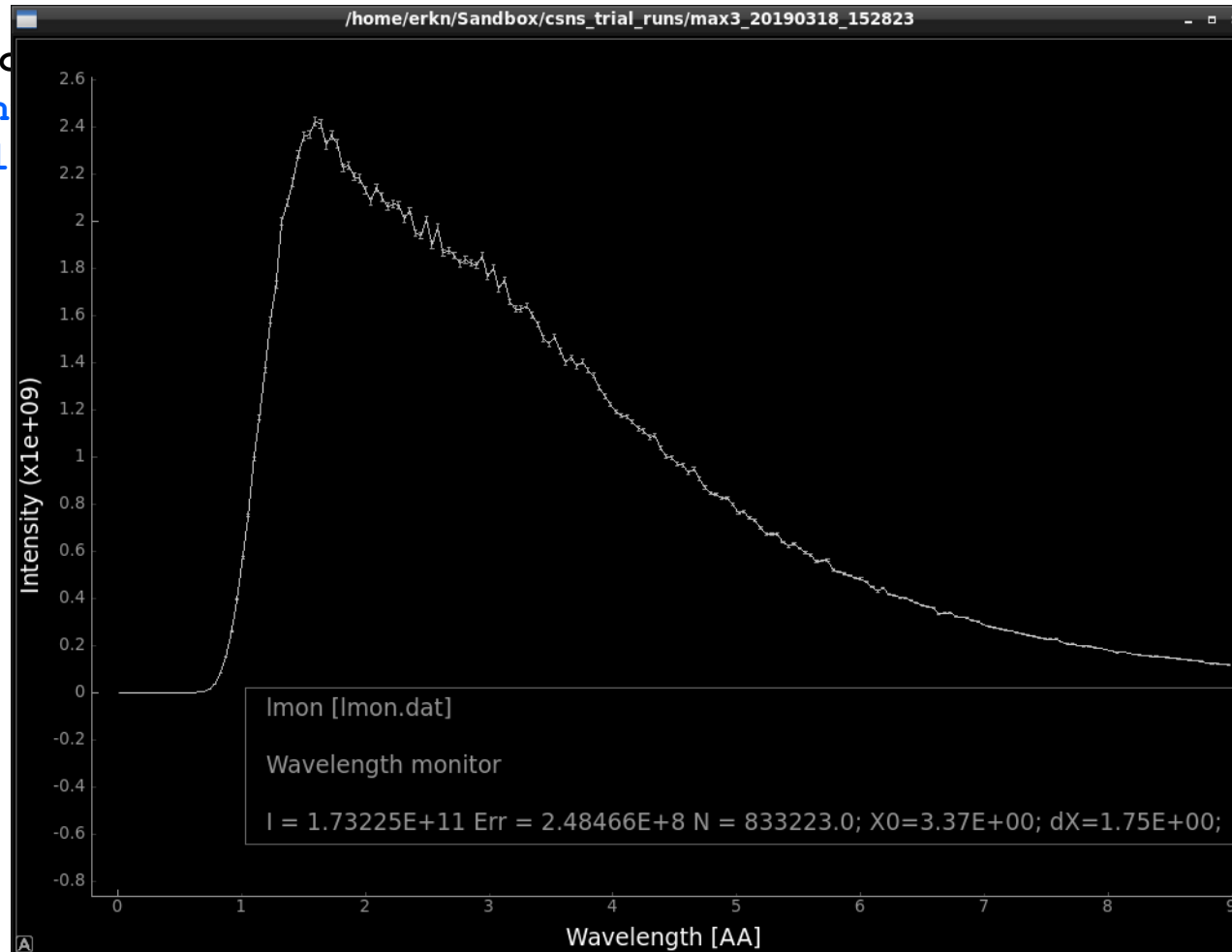
$$\alpha = 949.0 \text{ K } \text{\AA}^2 / T_i$$

Sources: Source_Maxwell_3

COMPONENT source

Lmin

T1=1



26,

yh = 0.12,

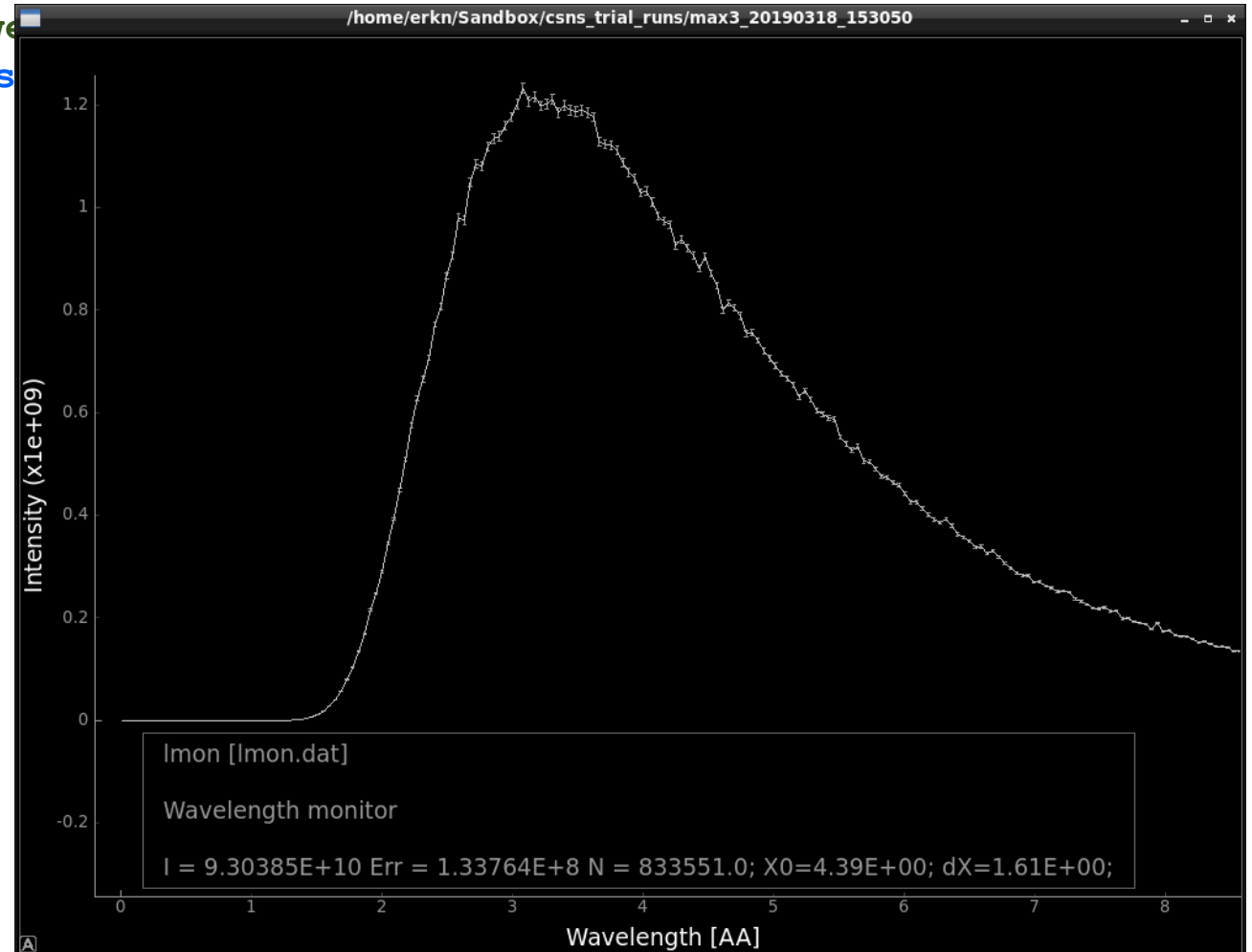
I3=0.95E11)

from the PSI cold source

Sources: Source_Maxwell_3

```
COMPONENT source = Source_Maxwell_3
  Lmin=0.1, Lmax=9.0, dist=150.42,
  T1=150.42, I1=3.67E11,
```

Just for fun – let's see what happens if we remove the fast peak...



Input parameters

McStas



Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
size	m	Edge of cube shaped source (for backward compatibility)	0
yheight	m	Height of rectangular source	0
xwidth	m	Width of rectangular source	0
Lmin	AA	Lower edge of lambda distribution	
Lmax	AA	Upper edge of lambda distribution	
dist	m	Distance from source to focusing rectangle; at (0,0,dist)	
focus_xw	m	Width of focusing rectangle	
focus_yh	m	Height of focusing rectangle	
T1	K	1st temperature of thermal distribution	
T2	K	2nd temperature of thermal distribution	300
T3	K	3rd temperature of - - -	300
I1	1/(cm**2*st)	flux, 1 (in flux units, see above)	
I2	1/(cm**2*st)	flux, 2 (in flux units, see above)	0
I3	1/(cm**2*st)	flux, 3 - - -	0
target_index	1	relative index of component to focus at, e.g. next is +1 this is used to compute 'dist' automatically.	+1
lambda0	AA	Mean wavelength of neutrons.	0
dlambda	AA	Wavelength spread of neutrons.	0

2025 ISIS
McStas
School



Sources: Source_gen (Source_gen4)

```
COMPONENT source = Source_gen(yheight=0.156, xwidth=0.126,  
                                Lmin=0.1, Lmax=9.0, dist=1.5, focus_xw = 0.025, focus_yh = 0.12,  
                                T1=150.42, I1=3.67E11, T2=38.74, I2=3.64E11, T3=14.84, I3=0.95E11)
```

Almost the same as Source_Maxwell_3: but with optional flux-files as input.



MCPL_input/output

Reads/writes events directly from MCPL-format files:
“T. Kittelmann et. al., “”, J. Phys. Comp., 2017

```
COMPONENT vout = MCPL_output(filename="voutput.mcpl",  
                                doubleprec=1,polarisationuse=1,verbose=1)  
AT (Xout,Yout,Zout) RELATIVE PREVIOUS
```

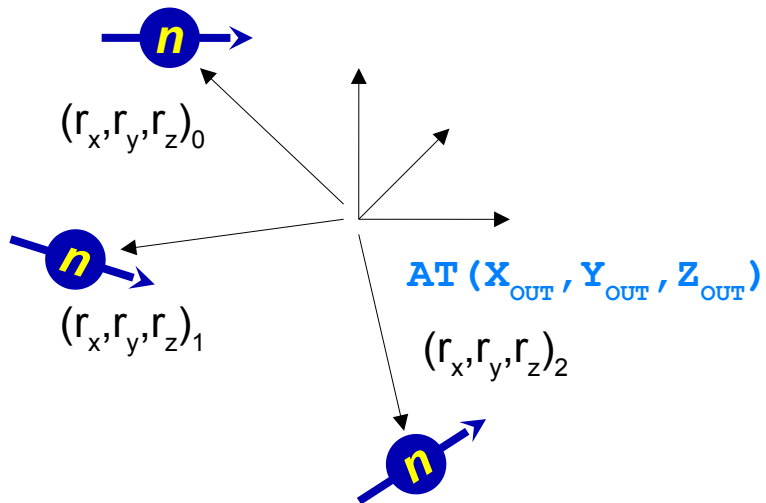
```
COMPONENT vin = MCPL_input(filename="voutput.mcpl",  
                              polarisationuse=1,verbose=1)  
AT (Xin,Yin,Zin) RELATIVE PREVIOUS
```



MCPL_input/output

Can include an Implicit Translation:

MCPL_output.comp

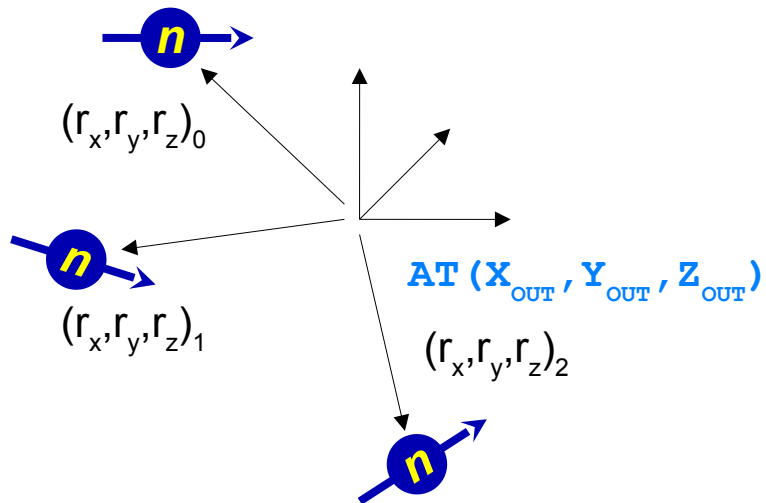




MCPL_input/output

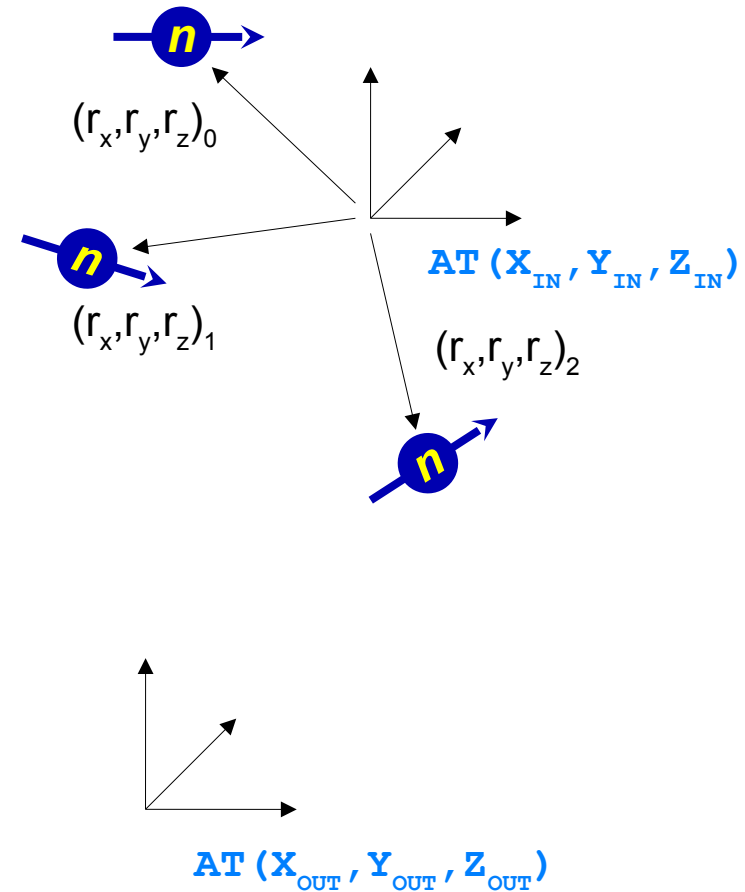
Can include an Implicit Translation:

MCPL_output.comp



BECOMES

MCPL_input.comp





Pulsed sources:

Simplest case:

Use a continuous source!

Model a source with given wavelength and spatial distribution

and

... an infinitely short pulse length. I.e. $t = 0$ for all neutron rays.

```
COMPONENT src = Source_simple(  
    radius=0.05, lambda0=2.5, dlambda=1.5,  
    focus_xw=0.1, focus_yh=0.1, dist=5 )  
AT(0,0,0) RELATIVE origin
```

2025 ISIS
McStas
School



Pulsed sources:

Simplest case:

Use a continuous source!

Model a source with given wavelength and spatial distribution

and

... an infinitely short pulse length. I.e. $t = 0$ for all neutron rays.

```
COMPONENT src = Source_simple(  
    radius=0.05, lambda0=2.5, dlambda=1.5,  
    focus_xw=0.1, focus_yh=0.1, dist=5 )  
AT(0,0,0) RELATIVE origin  
EXTEND  
% {  
    t=0;  
% }
```



Pulsed Sources:

Simplest case:

Use a continuous source!

Model a source with given wavelength and spatial distribution

... an infinitely sh

neutron rays.

```

COMPONENT s1
  radius=0.
  focus_xw=
  lambda=1.5,
  t=0.1, dist=5 )
AT (0,0,0) REL origin
EXTEND
%{
  t=0;
%}

```

Or: Use a chopper
(see later)



Pulsed Sources: Moderator

A flat pulsed source with uniform energy spectrum:

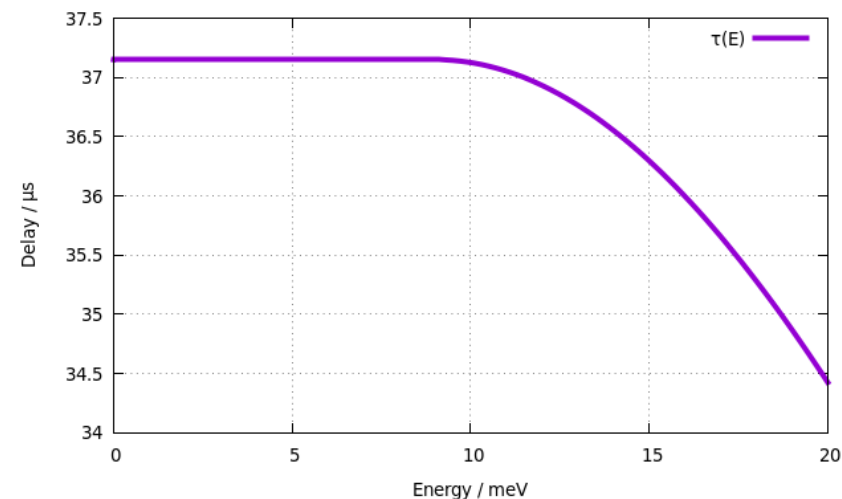
$$x \in U\left[-\frac{xwidth}{2}, \frac{xwidth}{2}\right] \quad y \in U\left[-\frac{yheight}{2}, \frac{yheight}{2}\right]$$

$$|v| = f(\lambda) ; \lambda \in U[L_{min} L_{max}]$$

Time structure is given by energy dependent probability density function:

$$f_t = \frac{1}{\tau} \exp\left(-\frac{t}{\tau}\right)$$

$$\tau = \begin{cases} t_0; & E < E_c \\ t_0 \left(\frac{1}{1 + \frac{(E - E_c)}{\gamma}} \right); & E \geq E_c \end{cases}$$





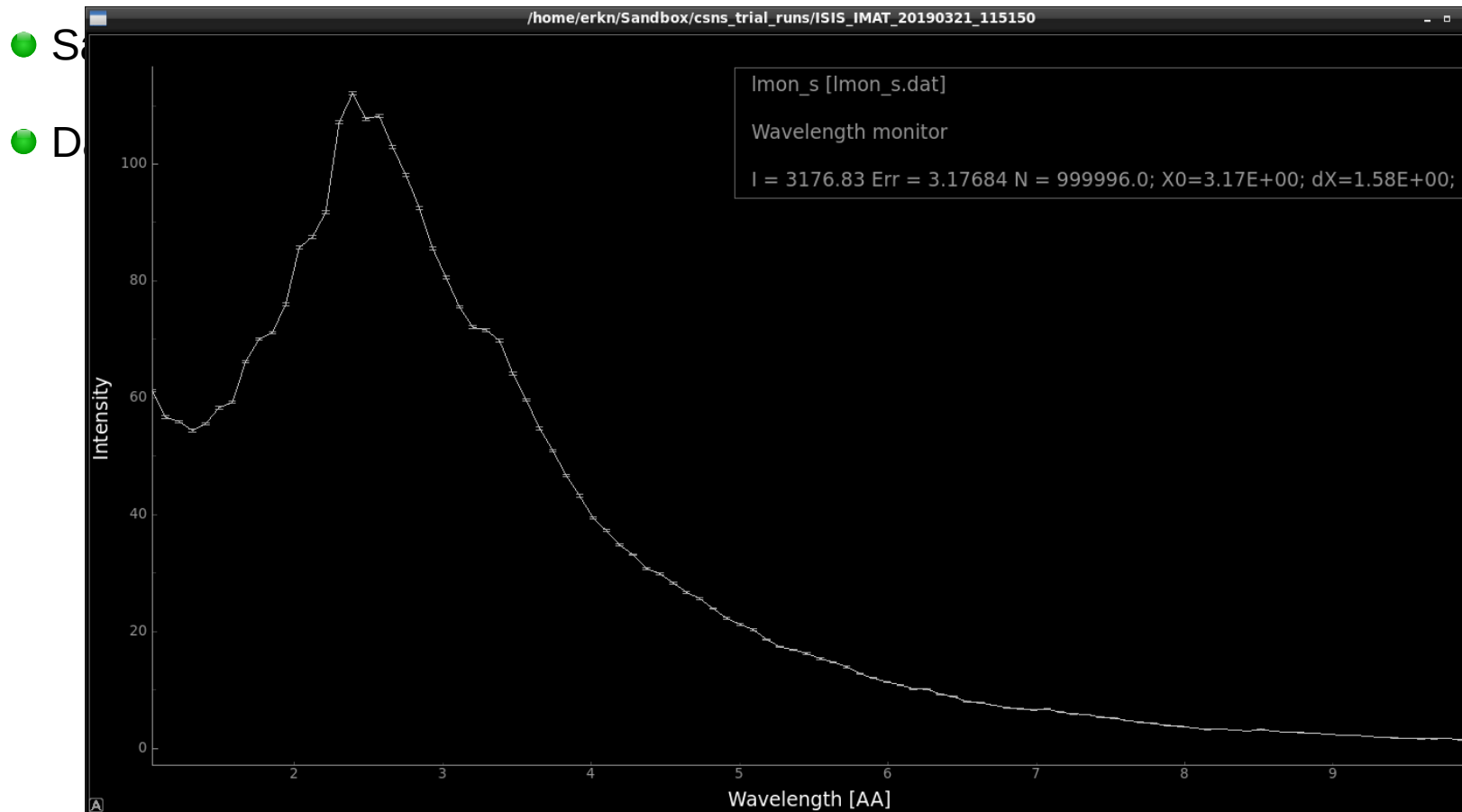
Pulsed Sources: ViewModISIS

- Samples directly from tallies coming from e.g. MCNP target+moderator calculations.
- Data file supplied for each beam port at ISIS.

2025 ISIS
McStas
School

Pulsed Sources: ViewModISIS

ISIS T2: IMAT



culations.

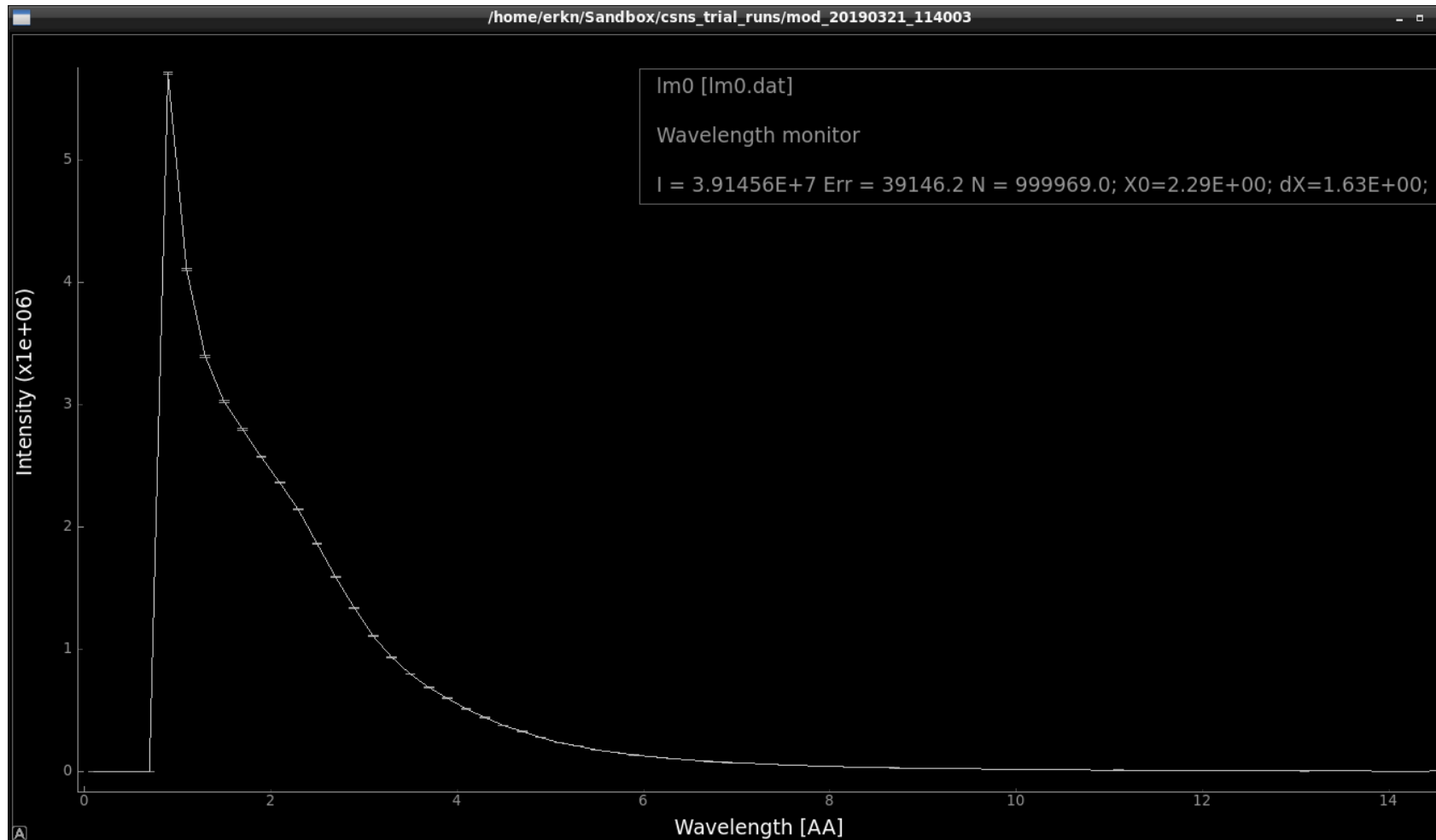


Pulsed Sources: SNS_source

- Samples directly from tallies coming from e.g. MCNP target+moderator calculations.
- Originally from SNS but also used extensively at J-PARC
- Can be used (with the proper input files) to model CSNS, and likely also ISIS.

2025 ISIS
McStas
School

Pulsed Sources: SNS_source



calculations.

McStas.



Pulsed Sources: SNS_source_analytic

- Samples from fits of Padé-functions to tallies from SNS_source.
 - Requires a complex fitting campaign
 - + Much faster than SNS_source
 - + “Cleaner” distributions where statistics are sketchy
- Can be used (with the proper input files) to model CSNS-source.

2025 ISIS
McStas
School



Monitors (some)

1D

- ◆ L_monitor $\rightarrow I(\lambda)$
- ◆ TOF_monitor $\rightarrow I(t)$
- ◆ Hdiv_monitor $\rightarrow I(\text{div}_x)$
- ◆ MeanPolLambda $\rightarrow \langle \bar{\mathbf{P}} \rangle(\lambda)$
- ◆ E_monitor $\rightarrow I(E)$

2D

- PSD_monitor $\rightarrow I(x, y)$
- PSD_monitor_4PI $\rightarrow I(\theta, \phi)$
- PolLambda_monitor $\rightarrow I(\bar{\mathbf{P}}, \lambda)$
- Divergence_monitor $\rightarrow I(\text{div}_x, \text{div}_y)$
- DivPos_monitor $\rightarrow I(\text{div}_x, x)$

nD

- Monitor_nD $\rightarrow I(X)$
- or
- $I(X, Y)$
- or
- $Z(X, Y, Z)$
- or ...



Monitors: Quick examples

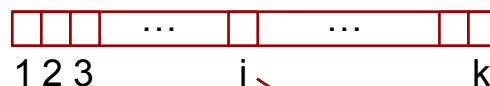
```
COMPONENT my_L_monitor = L_monitor(xwidth=0.2, yheight=0.2,  
                                     nL=20, filename="Output.L", Lmin=2, Lmax=10)
```



2025 ISIS
McStas
School

In a histogram sense

♦ Imagine a histogram, e.g. $\mathbf{I}(\lambda)$

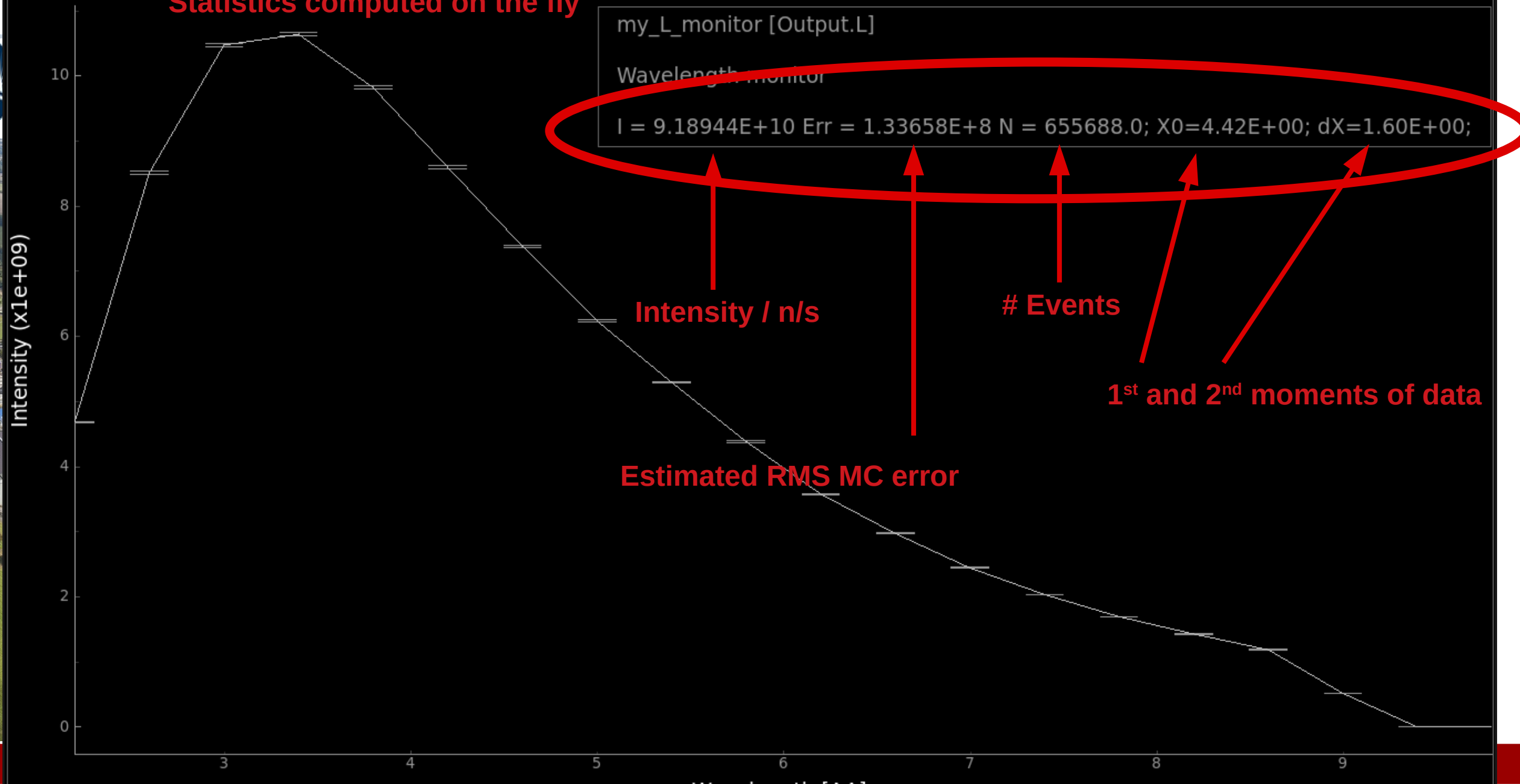


In bin i , $\mathbf{N}$ events each carrying a fractional intensity p_j so that

$$I = \sum_N p_j$$

♦ The RMS variance over that set becomes our statistical error bar $\mathbf{E}$

Statistics computed on the fly



Statistics computed on the fly

Intensity (x1e+09)

my_L_monitor [Output.L]

Wavelength monitor

$I = 9.18944\text{E}+10$ Err = $1.33658\text{E}+8$ N = 655688.0; X0=4.42E+00; dX=1.60E+00;

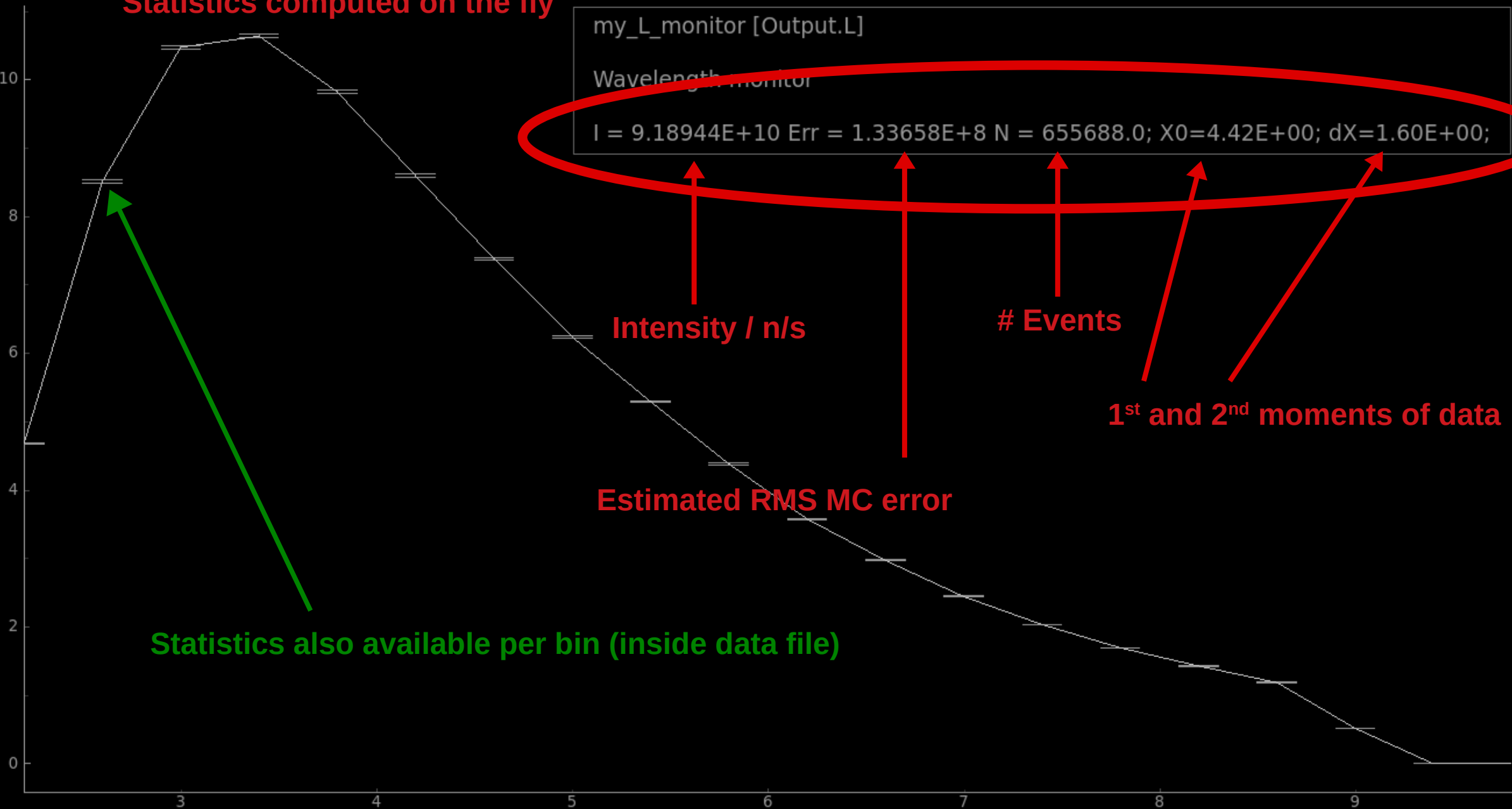
Intensity / n/s

Events

1st and 2nd moments of data

Estimated RMS MC error

Statistics also available per bin (inside data file)



He3H [He3.psd]

PSD monitor

$I = 9.67067\text{E-}10$ Err = $6.31266\text{E-}12$ N = 25846.0; X0= $6.19\text{E-}02$; dX= $4.03\text{E-}01$; Y0= $2.07\text{E-}02$; dY= $1.43\text{E+}00$;

In 2D

1st and 2nd moments

Y position [cm]

X position [cm]

From "Virtual experiments - the ultimate aim of neutron ray-tracing simulations", K. Lefmann et al., Journal of Neutron Research 16, 97-111 (2008)



Let n be the number of neutron rays reaching the detector, and let the rays have (different) weights, w_i . The simulated intensity is then given by

$$I = \sum_{i=1}^n w_i. \quad (1)$$

The estimate of the error on this number is calculated in the McStas manual [1], and the standard deviation is approximated by

$$\sigma^2(I) = \sum_{i=1}^n w_i^2. \quad (2)$$

In real experiments, $w_i = 1$, whence we reach $I = n$ and $\sigma(I) = \sqrt{I}$ as expected (for counts exceeding 10). Let the virtual time be denoted by t . The simulated counts during this time becomes

$$C = tI, \quad (3)$$

From "Virtual experiments - the ultimate aim of neutron ray-tracing simulations", K. Lefmann et al., Journal of Neutron Research 16, 97-111 (2008)



and its error bar estimate is

$$\sigma^2(C) = t^2 \sigma^2(I). \quad (4)$$

However, to simulate a realistic counting statistics, we must fulfill

$$\sigma_{\text{VE}}(C_{\text{VE}}) = \sqrt{C_{\text{VE}}}. \quad (5)$$

This is obtained by adding to (3) a Gaussian noise $E(\Sigma)$ of mean value zero and standard deviation Σ :

$$C_{\text{VE}} = tI + E(\Sigma). \quad (6)$$

The standard deviation for the VE becomes

$$\sigma_{\text{VE}}^2(C) = t^2 \sigma^2(I) + \Sigma^2. \quad (7)$$

Now, the requirement (5) allows us to determine Σ :

$$\Sigma^2 = tI - t^2 \sigma^2(I). \quad (8)$$

Since Σ^2 must remain positive, we reach an upper limit on t

$$t_{\text{max}} = \frac{I}{\sigma^2(I)}. \quad (9)$$



Sketch of an algorithm...

1. On a given McStas histogram
2. For the non-zero bins. calculate

$$t_{\max} = \frac{I}{\sigma^2(I)}.$$

$t_{\max}$

The *smallest* $t_{\max}$ defines the “maximal counting time” allowed by your statistics

3. Preferably a “background” should be added - use a “known experimental value” or an estimate...



Monitor_nD

The all-in-one , swiss-army-knife of monitors

***Monitor_nD** can have almost any shape, and record any requested standard quantities*



Provides foundation for event-processing in



2025 ISIS
McStas
School



Monitor_nD

A general monitor for 0D/1D/2D records

Examples

```
COMPONENT MyMon = Monitor_nD( xwidth = 0.1, yheight = 0.1, zdepth = 0,  
options = "intensity per cm2 angle,limits=[-5 5],  
bins=10,with borders, file = mon1")
```

options = "banana, theta limits=[10,130], bins=120, y"

options = "multiple kx ky kz, auto abs log t, and list all neutrons"



Monitor_nD

... or monitor just about anything:

```
COMPONENT MyMon = Monitor_nD(xwidth = 0.1, yheight = 0.1,  
    user1="age", username1="Age of the Captain [years]",  
    options="user1, auto")
```

2025 ISIS
McStas
School



Exercise 2:

Head over to the github site and continue the exercise we started before:

https://github.com/mccode-dev/Schools/tree/master/2025/ISIS_November_2025/1_Tuesday_November_4th/2_Component_basics/Exercise



**2025 ISIS
McStas
School**



EUROPEAN
SPALLATION
SOURCE



DTU

